CHAPTER

22

# Audio Processing

*Audio processing* covers many diverse fields, all involved in presenting sound to human listeners. Three areas are prominent: (1) *high fidelity music reproduction*, such as in audio compact discs, (2) *voice telecommunications*, another name for telephone networks, and (3) *synthetic speech*, where computers generate and recognize human voice patterns. While these applications have different goals and problems, they are linked by a common umpire: the human ear. Digital Signal Processing has produced revolutionary changes in these and other areas of audio processing.

## Human Hearing

The human ear is an exceedingly complex organ. To make matters even more difficult, the information from *two* ears is combined in a perplexing neural network, the human brain. Keep in mind that the following is only a brief overview; there are many subtle effects and poorly understood phenomena related to human hearing.

Figure 22-1 illustrates the major structures and processes that comprise the human ear. The *outer ear* is composed of two parts, the visible flap of skin and cartilage attached to the side of the head, and the *ear canal*, a tube about 0.5 cm in diameter extending about 3 cm into the head. These structures direct environmental sounds to the sensitive *middle and inner ear* organs located safely inside of the skull bones. Stretched across the end of the ear canal is a thin sheet of tissue called the *tympanic membrane* or *ear drum*. Sound waves striking the tympanic membrane cause it to vibrate. The middle ear is a set of small bones that transfer this vibration to the *cochlea* (inner ear) where it is converted to neural impulses. The cochlea is a liquid filled tube roughly 2 mm in diameter and 3 cm in length. Although shown straight in Fig. 22-1, the cochlea is curled up and looks like a small snail shell. In fact, *cochlea* is derived from the Greek word for *snail*.